Franka Emika Robot Control Comparison Report

Automatic Analysis

November 18, 2025

Contents

1 Overview

This report presents a comprehensive comparison of robot control performance across different:

- Controllers: FBL (Feedback Linearization) and PBC (Passivity-Based Control)
- Uncertainty levels: high, extreme
- Initial conditions: matched vs mismatched
- Gain tuning: optimized vs unoptimized
- Trajectory types: quintic, bang-bang, bang-coast-bang

Total simulations analyzed: 96

2 Controller Performance Summary

2.1 FBL Controller

Metric	Value
Number of simulations	48
Mean EE max error [mm]	34.67
Std EE max error [mm]	21.34
Best EE max error [mm]	10.15
Worst EE max error [mm]	83.23
Mean joint max error [deg]	13.63
Best joint max error [deg]	7.42
Worst joint max error [deg]	20.59

Table 1: Summary statistics for FBL controller

2.2 PBC Controller

Metric	Value
Number of simulations	48
Mean EE max error [mm]	14.12
Std EE max error [mm]	7.33
Best EE max error [mm]	2.40
Worst EE max error [mm]	20.84
Mean joint max error [deg]	5.71
Best joint max error [deg]	0.82
Worst joint max error [deg]	10.00

Table 2: Summary statistics for PBC controller

3 Detailed Performance Tables

Performance tables show results grouped by trajectory type, with test labels A and B, including averages for each configuration. Lower values indicate better performance.

3.1 Uncertainty Level: HIGH

3.1.1 Matched Initial Conditions, Optimized Gains

Table 3: FBL Controller - high uncertainty - Matched Initial Conditions, Optimized Gains

Trajectory	Test	RMS Error (rad)	RMS Error (deg)	Final EE Error (mm)
Quintic	A	0.0394	2.25	8.71
	B	0.0425	2.44	12.32
Bang-Bang	A	0.0367	2.10	4.83
	B	0.0419	2.40	12.15
Bang-Coast-Bang	A	0.0367	2.10	4.22
	B	0.0412	2.36	12.40
Average		**0.0397**	**2.28**	**9.10**

Table 4: PBC Controller - high uncertainty - Matched Initial Conditions, Optimized Gains

Trajectory	Test	RMS Error (rad)	RMS Error (deg)	Final EE Error (mm)
Quintic	A	0.0063	0.36	5.77
	B	0.0080	0.46	5.66
Bang-Bang	A	0.0063	0.36	6.01
	B	0.0079	0.46	5.78
Bang-Coast-Bang	A	0.0063	0.36	6.02
	B	0.0079	0.45	5.80
Average		**0.0071**	**0.41**	**5.84**

3.1.2 Matched Initial Conditions, Unoptimized Gains

Table 5: FBL Controller - high uncertainty - Matched Initial Conditions, Unoptimized Gains

Trajectory	Test	RMS Error (rad)	RMS Error (deg)	Final EE Error (mm)
Quintic	A	0.0933	5.35	12.06
	B	0.0898	5.15	35.40
Bang-Bang	A	0.0925	5.30	17.10
	B	0.0891	5.10	32.77
Bang-Coast-Bang	A	0.0929	5.32	16.98
	B	0.0885	5.07	32.75
Average		**0.0910**	**5.22**	**24.51**

Table 6: PBC Controller - high uncertainty - Matched Initial Conditions, Unoptimized Gains

Trajectory	Test	RMS Error (rad)	RMS Error (deg)	Final EE Error (mm)
Quintic	A	0.0078	0.45	6.78
	B	0.0099	0.57	6.63
Bang-Bang	A	0.0076	0.44	7.05
	B	0.0099	0.57	6.89
Bang-Coast-Bang	A	0.0076	0.44	7.08
	B	0.0099	0.57	6.89
Average		**0.0088**	**0.50**	**6.89**

3.1.3 Mismatched Initial Conditions, Optimized Gains

Table 7: FBL Controller - high uncertainty - Mismatched Initial Conditions, Optimized Gains

Trajectory	Test	RMS Error (rad)	RMS Error (deg)	Final EE Error (mm)
Quintic	A	0.0440	2.52	8.71
	B	0.0459	2.63	12.33
Bang-Bang	A	0.0425	2.44	4.82
	B	0.0464	2.66	12.15
Bang-Coast-Bang	A	0.0425	2.43	4.22
	B	0.0459	2.63	12.40
Average		**0.0445**	**2.55**	**9.11**

Table 8: PBC Controller - high uncertainty - Mismatched Initial Conditions, Optimized Gains

Trajectory	Test	RMS Error (rad)	RMS Error (deg)	Final EE Error (mm)
Quintic	A	0.0381	2.18	5.77
	B	0.0317	1.81	5.66
Bang-Bang	A	0.0379	2.17	6.01
	B	0.0300	1.72	5.78
Bang-Coast-Bang	A	0.0377	2.16	6.02
	B	0.0305	1.75	5.80
Average		**0.0343**	**1.97**	**5.84**

3.1.4 Mismatched Initial Conditions, Unoptimized Gains

Table 9: FBL Controller - high uncertainty - Mismatched Initial Conditions, Unoptimized Gains

Trajectory	Test	RMS Error (rad)	RMS Error (deg)	Final EE Error (mm)
Quintic	A	0.0932	5.34	15.29
	B	0.0908	5.20	35.39
Bang-Bang	A	0.0924	5.29	16.91
	B	0.0901	5.16	32.77
Bang-Coast-Bang	A	0.0927	5.31	16.70
	B	0.0901	5.16	32.75
Average		**0.0915**	**5.24**	**24.97**

Table 10: PBC Controller - high uncertainty - Mismatched Initial Conditions, Unoptimized Gains

Trajectory	Test	RMS Error (rad)	RMS Error (deg)	Final EE Error (mm)
Quintic	A	0.0320	1.84	6.78
	B	0.0267	1.53	6.63
Bang-Bang	A	0.0351	2.01	7.05
	B	0.0271	1.55	6.89
Bang-Coast-Bang	A	0.0350	2.00	7.08
	B	0.0259	1.48	6.90
Average		**0.0303**	**1.74**	**6.89**

3.2 Uncertainty Level: EXTREME

3.2.1 Matched Initial Conditions, Optimized Gains

Table 11: FBL Controller - extreme uncertainty - Matched Initial Conditions, Optimized Gains

Trajectory	Test	RMS Error (rad)	RMS Error (deg)	Final EE Error (mm)
Quintic	A	0.0524	3.00	14.34
	B	0.0532	3.05	37.83
Bang-Bang	A	0.0483	2.77	15.42
	B	0.0515	2.95	31.87
Bang-Coast-Bang	A	0.0483	2.77	14.92
	B	0.0505	2.89	31.64
Average		**0.0507**	**2.90**	**24.34**

Table 12: PBC Controller - extreme uncertainty - Matched Initial Conditions, Optimized Gains

Trajectory	Test	RMS Error (rad)	RMS Error (deg)	Final EE Error (mm)
Quintic	A	0.0069	0.40	2.35
	B	0.0104	0.59	8.66
Bang-Bang	A	0.0069	0.40	2.39
	B	0.0103	0.59	8.82
Bang-Coast-Bang	A	0.0069	0.40	2.39
	B	0.0103	0.59	8.84
Average		**0.0086**	**0.49**	**5.58**

3.2.2 Matched Initial Conditions, Unoptimized Gains

Table 13: FBL Controller - extreme uncertainty - Matched Initial Conditions, Unoptimized Gains

Trajectory	Test	RMS Error (rad)	RMS Error (deg)	Final EE Error (mm)
Quintic	A	0.1187	6.80	30.65
	B	0.1168	6.69	80.51
Bang-Bang	A	0.1186	6.79	30.99
	B	0.1161	6.65	82.66
Bang-Coast-Bang	A	0.1176	6.74	30.86
	B	0.1143	6.55	81.74
Average		**0.1170**	**6.70**	**56.23**

Table 14: PBC Controller - extreme uncertainty - Matched Initial Conditions, Unoptimized Gains

Trajectory	Test	RMS Error (rad)	RMS Error (deg)	Final EE Error (mm)
Quintic	A	0.0086	0.49	2.76
	B	0.0126	0.72	10.12
Bang-Bang	A	0.0084	0.48	2.84
	B	0.0125	0.72	10.38
Bang-Coast-Bang	A	0.0084	0.48	2.84
	B	0.0124	0.71	10.42
Average		**0.0105**	**0.60**	**6.56**

3.2.3 Mismatched Initial Conditions, Optimized Gains

Table 15: FBL Controller - extreme uncertainty - Mismatched Initial Conditions, Optimized Gains

Trajectory	Test	RMS Error (rad)	RMS Error (deg)	Final EE Error (mm)
Quintic	A	0.0541	3.10	14.34
	B	0.0554	3.17	37.83
Bang-Bang	A	0.0513	2.94	15.42
	B	0.0535	3.06	31.87
Bang-Coast-Bang	A	0.0512	2.93	14.92
	B	0.0526	3.02	31.64
Average		**0.0530**	**3.04**	**24.34**

Table 16: PBC Controller - extreme uncertainty - Mismatched Initial Conditions, Optimized Gains

Trajectory	Test	RMS Error (rad)	RMS Error (deg)	Final EE Error (mm)
Quintic	A	0.0375	2.15	2.35
	B	0.0336	1.93	8.66
Bang-Bang	A	0.0366	2.10	2.39
	B	0.0324	1.86	8.82
Bang-Coast-Bang	A	0.0369	2.12	2.39
	B	0.0331	1.90	8.84
Average		**0.0350**	**2.01**	**5.58**

3.2.4 Mismatched Initial Conditions, Unoptimized Gains

Table 17: FBL Controller - extreme uncertainty - Mismatched Initial Conditions, Unoptimized Gains

Trajectory	Test	RMS Error (rad)	RMS Error (deg)	Final EE Error (mm)
Quintic	A	0.1184	6.78	34.02
	B	0.1157	6.63	64.76
Bang-Bang	A	0.1180	6.76	34.41
	B	0.1153	6.61	66.10
Bang-Coast-Bang	A	0.1170	6.71	34.27
	B	0.1143	6.55	66.40
Average		**0.1164**	**6.67**	**49.99**

Table 18: PBC Controller - extreme uncertainty - Mismatched Initial Conditions, Unoptimized Gains

Trajectory	Test	RMS Error (rad)	RMS Error (deg)	Final EE Error (mm)
Quintic	A	0.0327	1.87	2.76
	B	0.0292	1.67	10.12
Bang-Bang	A	0.0355	2.03	2.84
	B	0.0304	1.74	10.38
Bang-Coast-Bang	A	0.0354	2.03	2.84
	B	0.0304	1.74	10.42
Average		**0.0323**	**1.85**	**6.56**

4 FBL vs PBC Controller Comparison

This section provides direct head-to-head comparisons between FBL and PBC controllers under identical conditions.

4.1 Uncertainty Level: HIGH

4.1.1 Matched Initial Conditions, Optimized Gains

Table 19: FBL vs PBC Comparison - high uncertainty - Matched Initial Conditions, Optimized Gains

Trajectory	Controller	Test	RMS Error (rad)	RMS Error (deg)	Final EE Error (mm)
Quintic	FBL	A	0.0394	2.25	8.71
	PBC	A	0.0063	0.36	5.77
	FBL	B	0.0425	2.44	12.32
	PBC	B	0.0080	0.46	5.66
Bang-Bang	FBL	A	0.0367	2.10	4.83
	PBC	A	0.0063	0.36	6.01
	FBL	B	0.0419	2.40	12.15
	PBC	B	0.0079	0.46	5.78
Bang-Coast-Bang	FBL	A	0.0367	2.10	4.22
	PBC	A	0.0063	0.36	6.02
	FBL	B	0.0412	2.36	12.40
	PBC	B	0.0079	0.45	5.80

4.1.2 Matched Initial Conditions, Unoptimized Gains

Table 20: FBL vs PBC Comparison - high uncertainty - Matched Initial Conditions, Unoptimized Gains

Trajectory	Controller	Test	RMS Error (rad)	RMS Error (deg)	Final EE Error (mm)
Quintic	FBL	A	0.0933	5.35	12.06
	PBC	A	0.0078	0.45	6.78
	FBL	B	0.0898	5.15	35.40
	PBC	B	0.0099	0.57	6.63
Bang-Bang	FBL	A	0.0925	5.30	17.10
	PBC	A	0.0076	0.44	7.05
	FBL	B	0.0891	5.10	32.77
	PBC	B	0.0099	0.57	6.89
Bang-Coast-Bang	FBL	A	0.0929	5.32	16.98
	PBC	A	0.0076	0.44	7.08
	FBL	B	0.0885	5.07	32.75
	PBC	B	0.0099	0.57	6.89

4.1.3 Mismatched Initial Conditions, Optimized Gains

Table 21: FBL vs PBC Comparison - high uncertainty - Mismatched Initial Conditions, Optimized Gains

Trajectory	Controller	Test	RMS Error (rad)	RMS Error (deg)	Final EE Error (mm)
Quintic	FBL	A	0.0440	2.52	8.71
	PBC	A	0.0381	2.18	5.77
	FBL	B	0.0459	2.63	12.33
	PBC	B	0.0317	1.81	5.66
Bang-Bang	FBL	A	0.0425	2.44	4.82
	PBC	A	0.0379	2.17	6.01
	FBL	B	0.0464	2.66	12.15
	PBC	B	0.0300	1.72	5.78
Bang-Coast-Bang	FBL	A	0.0425	2.43	4.22
	PBC	A	0.0377	2.16	6.02
	FBL	B	0.0459	2.63	12.40
	PBC	B	0.0305	1.75	5.80

4.1.4 Mismatched Initial Conditions, Unoptimized Gains

Table 22: FBL vs PBC Comparison - high uncertainty - Mismatched Initial Conditions, Unoptimized Gains

Trajectory	Controller	Test	RMS Error (rad)	RMS Error (deg)	Final EE Error (mm)
Quintic	FBL	A	0.0932	5.34	15.29
	PBC	A	0.0320	1.84	6.78
	FBL	B	0.0908	5.20	35.39
	PBC	B	0.0267	1.53	6.63
Bang-Bang	FBL	A	0.0924	5.29	16.91
	PBC	A	0.0351	2.01	7.05
	FBL	B	0.0901	5.16	32.77
	PBC	B	0.0271	1.55	6.89
Bang-Coast-Bang	FBL	A	0.0927	5.31	16.70
	PBC	A	0.0350	2.00	7.08
	FBL	B	0.0901	5.16	32.75
	PBC	B	0.0259	1.48	6.90

4.2 Uncertainty Level: EXTREME

4.2.1 Matched Initial Conditions, Optimized Gains

Table 23: FBL vs PBC Comparison - extreme uncertainty - Matched Initial Conditions, Optimized Gains

Trajectory	Controller	Test	RMS Error (rad)	RMS Error (deg)	Final EE Error (mm)
Quintic	FBL	A	0.0524	3.00	14.34
	PBC	A	0.0069	0.40	2.35
	FBL	B	0.0532	3.05	37.83
	PBC	B	0.0104	0.59	8.66
Bang-Bang	FBL	A	0.0483	2.77	15.42
	PBC	A	0.0069	0.40	2.39
	FBL	B	0.0515	2.95	31.87
	PBC	B	0.0103	0.59	8.82
Bang-Coast-Bang	FBL	A	0.0483	2.77	14.92
	PBC	A	0.0069	0.40	2.39
	FBL	B	0.0505	2.89	31.64
	PBC	B	0.0103	0.59	8.84

4.2.2 Matched Initial Conditions, Unoptimized Gains

Table 24: FBL vs PBC Comparison - extreme uncertainty - Matched Initial Conditions, Unoptimized Gains

Trajectory	Controller	Test	RMS Error (rad)	RMS Error (deg)	Final EE Error (mm)
Quintic	FBL	A	0.1187	6.80	30.65
	PBC	A	0.0086	0.49	2.76
	FBL	B	0.1168	6.69	80.51
	PBC	B	0.0126	0.72	10.12
Bang-Bang	FBL	A	0.1186	6.79	30.99
	PBC	A	0.0084	0.48	2.84
	FBL	B	0.1161	6.65	82.66
	PBC	B	0.0125	0.72	10.38
Bang-Coast-Bang	FBL	A	0.1176	6.74	30.86
	PBC	A	0.0084	0.48	2.84
	FBL	B	0.1143	6.55	81.74
	PBC	B	0.0124	0.71	10.42

4.2.3 Mismatched Initial Conditions, Optimized Gains

Table 25: FBL vs PBC Comparison - extreme uncertainty - Mismatched Initial Conditions, Optimized Gains

Trajectory	Controller	Test	RMS Error (rad)	RMS Error (deg)	Final EE Error (mm)
Quintic	FBL	A	0.0541	3.10	14.34
	PBC	A	0.0375	2.15	2.35
	FBL	B	0.0554	3.17	37.83
	PBC	B	0.0336	1.93	8.66
Bang-Bang	FBL	A	0.0513	2.94	15.42
	PBC	A	0.0366	2.10	2.39
	FBL	B	0.0535	3.06	31.87
	PBC	B	0.0324	1.86	8.82
Bang-Coast-Bang	FBL	A	0.0512	2.93	14.92
	PBC	A	0.0369	2.12	2.39
	FBL	B	0.0526	3.02	31.64
	PBC	B	0.0331	1.90	8.84

4.2.4 Mismatched Initial Conditions, Unoptimized Gains

Table 26: FBL vs PBC Comparison - extreme uncertainty - Mismatched Initial Conditions, Unoptimized Gains

Trajectory	Controller	Test	RMS Error (rad)	RMS Error (deg)	Final EE Error (mm)
Quintic	FBL	A	0.1184	6.78	34.02
	PBC	A	0.0327	1.87	2.76
	FBL	B	0.1157	6.63	64.76
	PBC	B	0.0292	1.67	10.12
Bang-Bang	FBL	A	0.1180	6.76	34.41
	PBC	A	0.0355	2.03	2.84
	FBL	B	0.1153	6.61	66.10
	PBC	B	0.0304	1.74	10.38
Bang-Coast-Bang	FBL	A	0.1170	6.71	34.27
	PBC	A	0.0354	2.03	2.84
	FBL	B	0.1143	6.55	66.40
	PBC	B	0.0304	1.74	10.42

5 Extreme Cases Analysis

5.1 Best Performance

Configuration:

- Controller: PBC
- Uncertainty: extreme
- Initial condition: matched
- Gains: optimized
- Trajectory: bang_coast_bang (label A)

Metrics:

- EE max error: 2.40 mm
- EE RMS error: 1.99 mm
- Joint max error: 0.82 deg
- Joint RMS error: 0.40 deg

5.2 Worst Performance

Configuration:

- Controller: FBL
- Uncertainty: extreme
- Initial condition: matched
- Gains: unoptimized
- Trajectory: bang_bang (label B)

Metrics:

- EE max error: 83.23 mm
- EE RMS error: 50.97 mm
- Joint max error: 17.23 deg
- Joint RMS error: 6.65 deg

6 Summary Comparison Table

This section presents a condensed head-to-head comparison using End-Effector Max Error [mm]. The better value is Green.

Table 27: Head-to-head controller comparison (lower is better)

Configuration	FBL EE Error [mm]	PBC EE Error [mm]	Difference [mm]
BB/extreme/Match-Opt (A)	19.05	2.41	16.64
BB/extreme/Match-Opt (B)	31.87	11.67	20.20
BCB/extreme/Match-Opt (A)	19.02	2.40	16.62
BCB/extreme/Match-Opt (B)	31.64	11.74	19.90
Quintic/extreme/Match-Opt (A)	19.10	2.41	16.69
Quintic/extreme/Match-Opt (B)	37.83	11.63	26.20
BB/extreme/Match-Unopt (A)	39.27	2.88	36.39
BB/extreme/Match-Unopt (B)	83.23	14.60	68.63
BCB/extreme/Match-Unopt (A)	39.23	2.88	36.35
BCB/extreme/Match-Unopt (B)	82.70	14.55	68.15
Quintic/extreme/Match-Unopt (A)	39.80	2.89	36.92
Quintic/extreme/Match-Unopt (B)	81.15	14.56	66.59
BB/extreme/Mismatch-Opt (A)	20.84	20.84	0.00
BB/extreme/Mismatch-Opt (B)	31.87	20.84	11.02
BCB/extreme/Mismatch-Opt (A)	20.84	20.84	0.00
BCB/extreme/Mismatch-Opt (B)	31.64	20.84	10.80
Quintic/extreme/Mismatch-Opt (A)	20.84	20.84	0.00

Configuration	FBL EE Error [mm]	PBC EE Error [mm]	Difference [mm]
Quintic/extreme/Mismatch-Opt (B)	37.83	20.84	16.98
BB/extreme/Mismatch-Unopt (A)	39.07	20.84	18.22
BB/extreme/Mismatch-Unopt (B)	81.44	20.84	60.60
BCB/extreme/Mismatch-Unopt (A)	39.05	20.84	18.21
BCB/extreme/Mismatch-Unopt (B)	81.17	20.84	60.32
Quintic/extreme/Mismatch-Unopt (A)	39.52	20.84	18.67
Quintic/extreme/Mismatch-Unopt (B)	82.13	20.84	61.28
BB/high/Match-Opt (A)	10.17	6.10	4.07
BB/high/Match-Opt (B)	14.04	6.38	7.66
BCB/high/Match-Opt (A)	10.15	6.09	4.06
BCB/high/Match-Opt (B)	13.68	6.38	7.31
Quintic/high/Match-Opt (A)	10.19	6.10	4.09
Quintic/high/Match-Opt (B)	14.54	6.36	8.18
BB/high/Match-Unopt (A)	21.04	7.31	13.74
BB/high/Match-Unopt (B)	48.51	7.89	40.61
BCB/high/Match-Unopt (A)	21.02	7.30	13.72
BCB/high/Match-Unopt (B)	48.56	7.85	40.71
Quintic/high/Match-Unopt (A)	20.93	7.31	13.62
Quintic/high/Match-Unopt (B)	48.28	7.87	40.41
BB/high/Mismatch-Opt (A)	20.84	20.84	0.00
BB/high/Mismatch-Opt (B)	20.84	20.84	0.00
BCB/high/Mismatch-Opt (A)	20.84	20.84	0.00
BCB/high/Mismatch-Opt (B)	20.84	20.84	0.00
Quintic/high/Mismatch-Opt (A)	20.84	20.84	0.00
Quintic/high/Mismatch-Opt (B)	20.84	20.84	0.00
BB/high/Mismatch-Unopt (A)	20.84	20.84	0.00
BB/high/Mismatch-Unopt (B)	48.51	20.84	27.67
BCB/high/Mismatch-Unopt (A)	20.84	20.84	0.00

Configuration	FBL EE Error [mm]	PBC EE Error [mm]	Difference [mm]
BCB/high/Mismatch-Unopt (B)	48.56	20.84	27.72
Quintic/high/Mismatch-Unopt (A)	20.99	20.84	0.15
Quintic/high/Mismatch-Unopt (B)	48.26	20.84	27.42

6.1 Overall Winner Statistics

- FBL wins: 11 (22.9%)
- PBC wins: 37 (77.1%)
- Total comparisons: 48